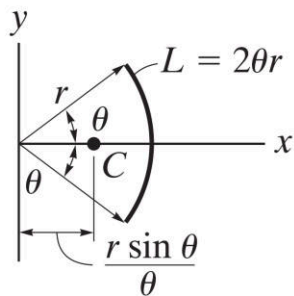


Gebze Technical University
Dep. of Mechanical Engineering

Statics (ME 231)
Lecture 18: Maximum
and Minimum Moments of Inertia
By Saeed Lotfan

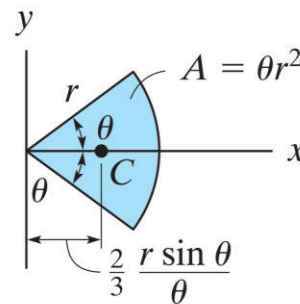
Geometric Properties of Line and Area Elements

Centroid Location



Circular arc segment

Centroid Location

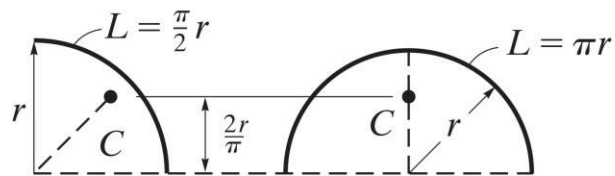


Circular sector area

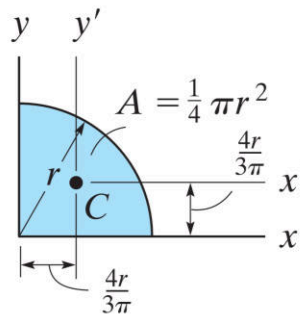
Area Moment of Inertia

$$I_x = \frac{1}{4} r^4 \left(\theta - \frac{1}{2} \sin 2\theta \right)$$

$$I_y = \frac{1}{4} r^4 \left(\theta + \frac{1}{2} \sin 2\theta \right)$$



Quarter and semicircle arcs



Quarter circle area

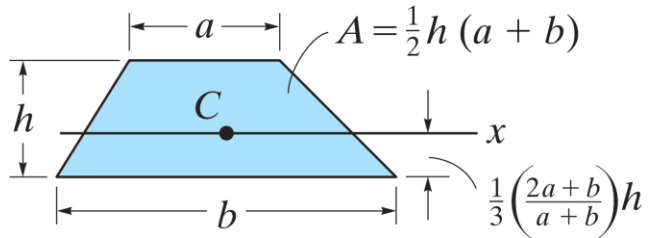
$$I_x = \frac{1}{16} \pi r^4$$

$$I_y = \frac{1}{16} \pi r^4$$

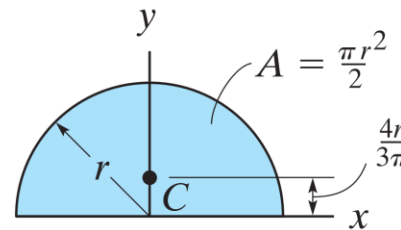
$$I_{x'} = \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) r^4$$

$$I_{y'} = \left(\frac{\pi}{16} - \frac{4}{9\pi} \right) r^4$$

Geometric Properties of Line and Area Elements



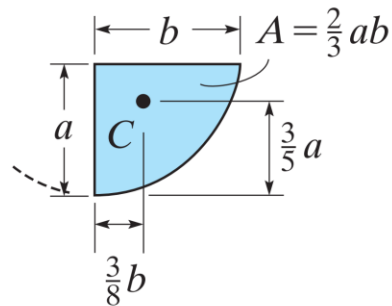
Trapezoidal area



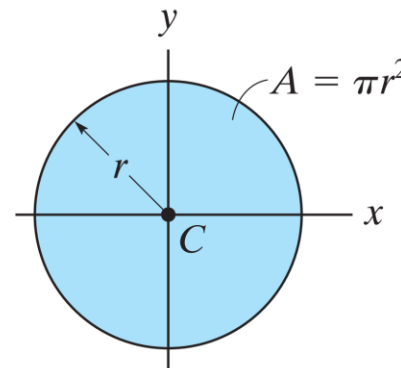
Semicircular area

$$I_x = \frac{1}{8} \pi r^4$$

$$I_y = \frac{1}{8} \pi r^4$$



Semiparabolic area

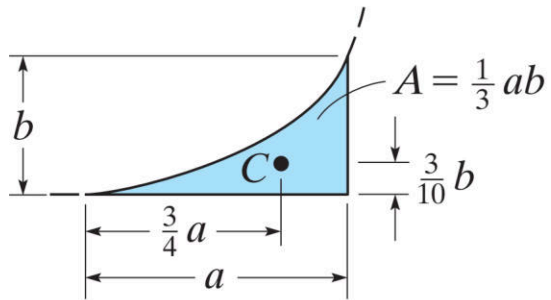


Circular area

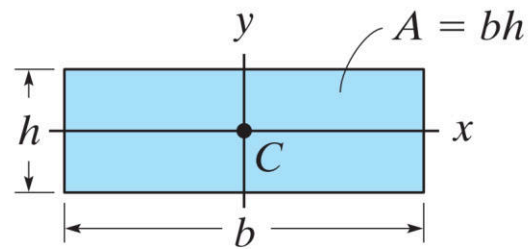
$$I_x = \frac{1}{4} \pi r^4$$

$$I_y = \frac{1}{4} \pi r^4$$

Geometric Properties of Line and Area Elements



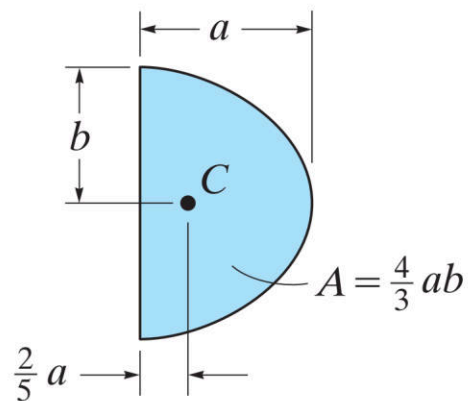
Exparabolic area



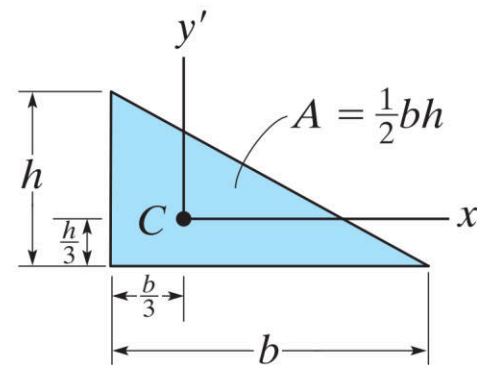
Rectangular area

$$I_x = \frac{1}{12} b h^3$$

$$I_y = \frac{1}{12} h b^3$$



Parabolic area



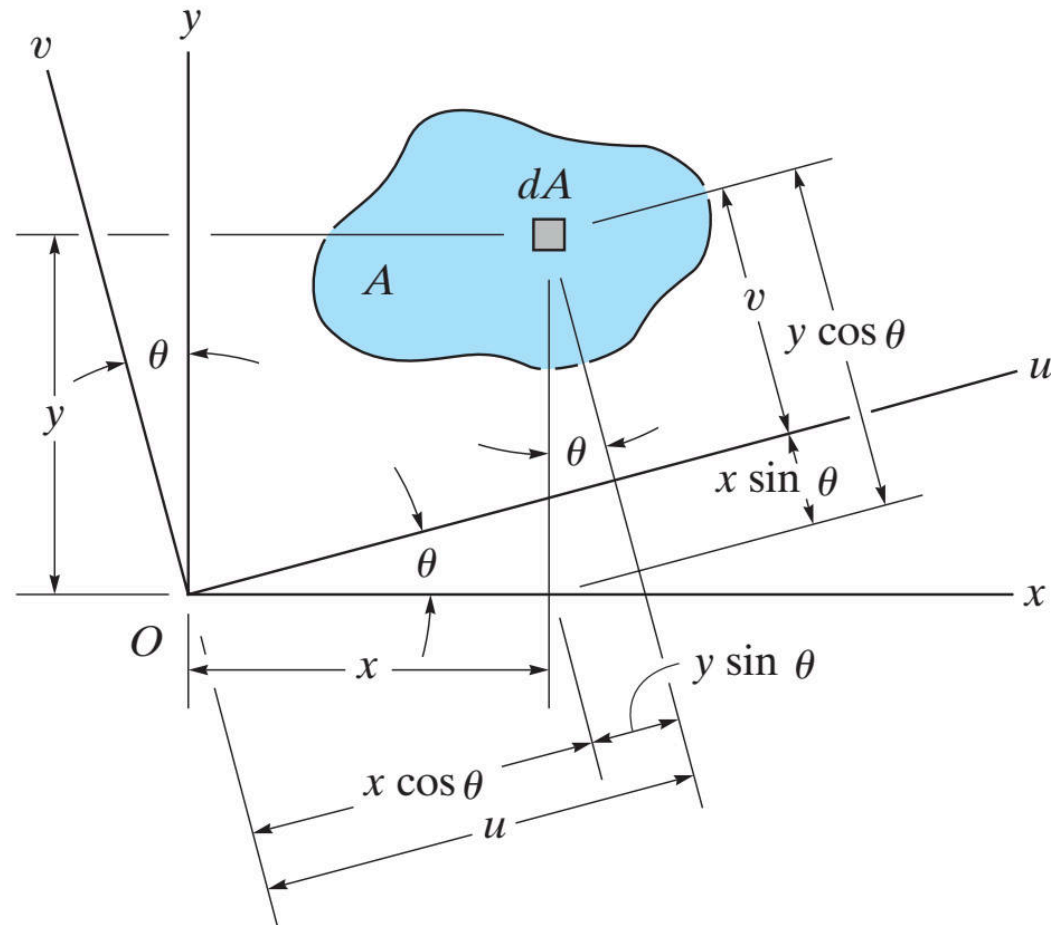
Triangular area

$$I_{x'} = \frac{1}{36} b h^3$$

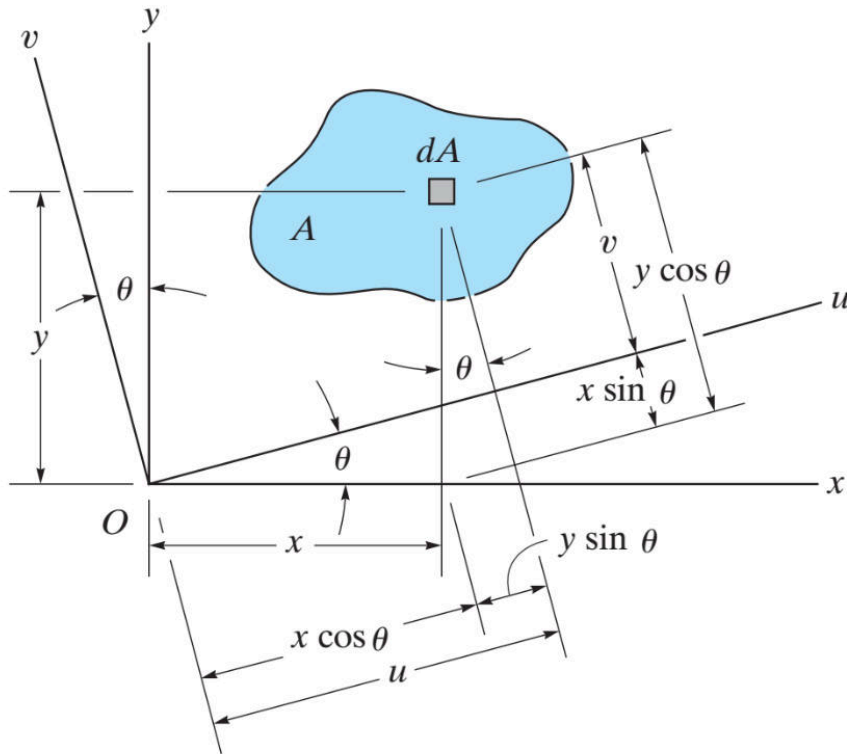
$$I_{y'} = \frac{1}{36} h b^3$$

Moments of Inertia and Different Loadings

Moments of Inertia for an Area About Inclined Axes



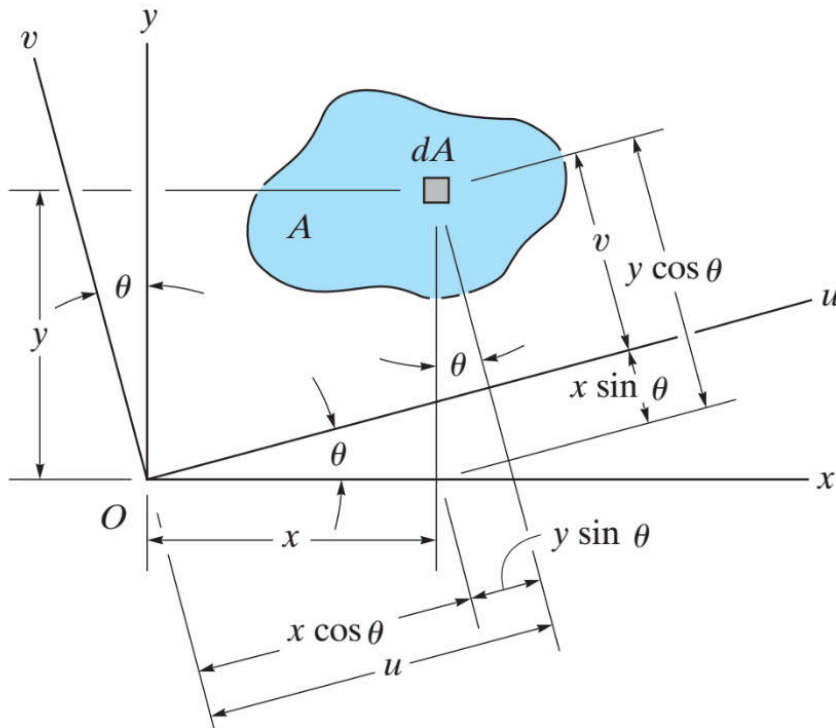
Moments of Inertia for an Area About Inclined Axes



$$u = x \cos \theta + y \sin \theta$$

$$v = y \cos \theta - x \sin \theta$$

Moments of Inertia for an Area About Inclined Axes



$$dI_u = v^2 dA = (y \cos \theta - x \sin \theta)^2 dA$$

$$dI_v = u^2 dA = (x \cos \theta + y \sin \theta)^2 dA$$

$$dI_{uv} = uv dA = (x \cos \theta + y \sin \theta)(y \cos \theta - x \sin \theta) dA$$

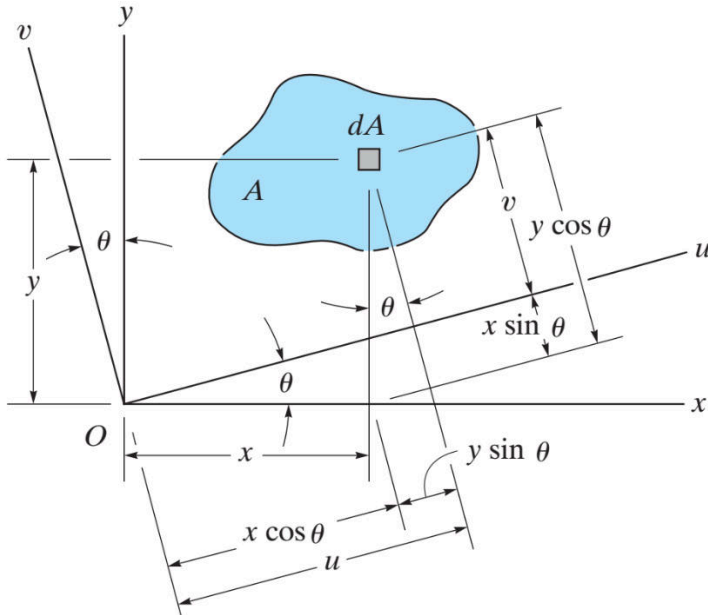
Moments of Inertia for an Area About Inclined Axes

$$I_u = I_x \cos^2 \theta + I_y \sin^2 \theta - 2I_{xy} \sin \theta \cos \theta$$

$$I_v = I_x \sin^2 \theta + I_y \cos^2 \theta + 2I_{xy} \sin \theta \cos \theta$$

$$I_{uv} = I_x \sin \theta \cos \theta - I_y \sin \theta \cos \theta + I_{xy}(\cos^2 \theta - \sin^2 \theta)$$

Moments of Inertia for an Area About Inclined Axes

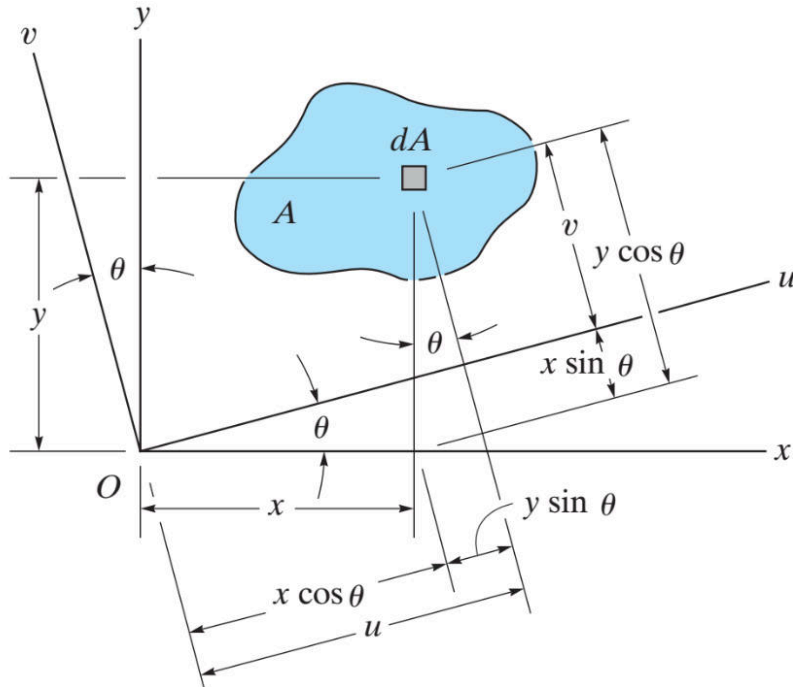


$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

Moments of Inertia for an Area About Inclined Axes



$$J_O = I_u + I_v = I_x + I_y$$

Moments of Inertia for an Area About Inclined Axes

$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$\frac{dI_u}{d\theta} = -2 \left(\frac{I_x - I_y}{2} \right) \sin 2\theta - 2I_{xy} \cos 2\theta = 0$$

$$\tan 2\theta_p = \frac{-I_{xy}}{(I_x - I_y)/2}$$

What happens to moments of inertia for area at the angle θ_p ?

Moments of Inertia for an Area About Inclined Axes

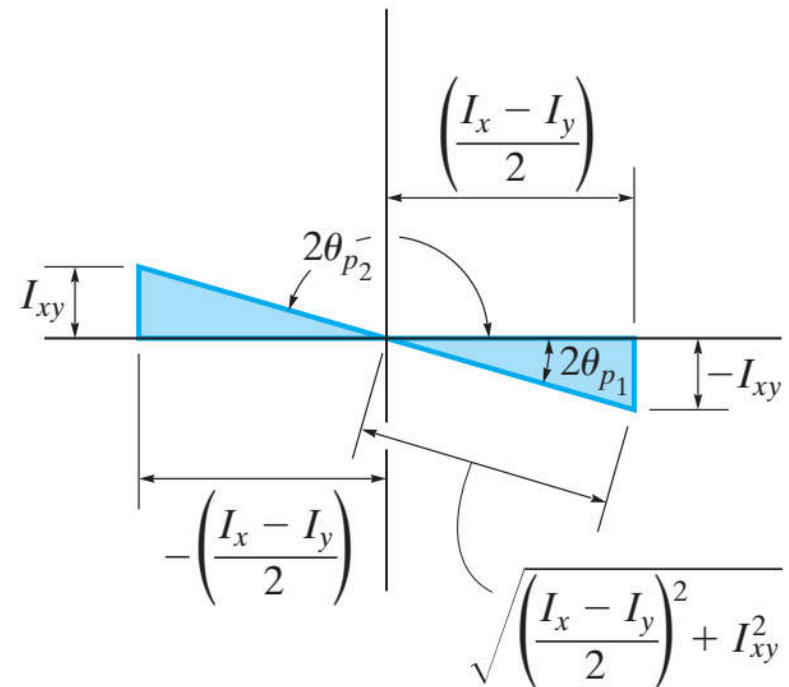
$$\tan 2\theta_p = \frac{-I_{xy}}{(I_x - I_y)/2}$$

$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

$$I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

$$I_{\max/\min} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$



Principal Moments of Inertia

$$\tan 2\theta_p = \frac{-I_{xy}}{(I_x - I_y)/2}$$

$$I_{\min}^{\max} = \frac{I_x + I_y}{2} \pm \sqrt{\left(\frac{I_x - I_y}{2}\right)^2 + I_{xy}^2}$$

Mohr's Circle for Moments of Inertia

$$I_u = \frac{I_x + I_y}{2} + \frac{I_x - I_y}{2} \cos 2\theta - I_{xy} \sin 2\theta$$

$$I_v = \frac{I_x + I_y}{2} - \frac{I_x - I_y}{2} \cos 2\theta + I_{xy} \sin 2\theta$$

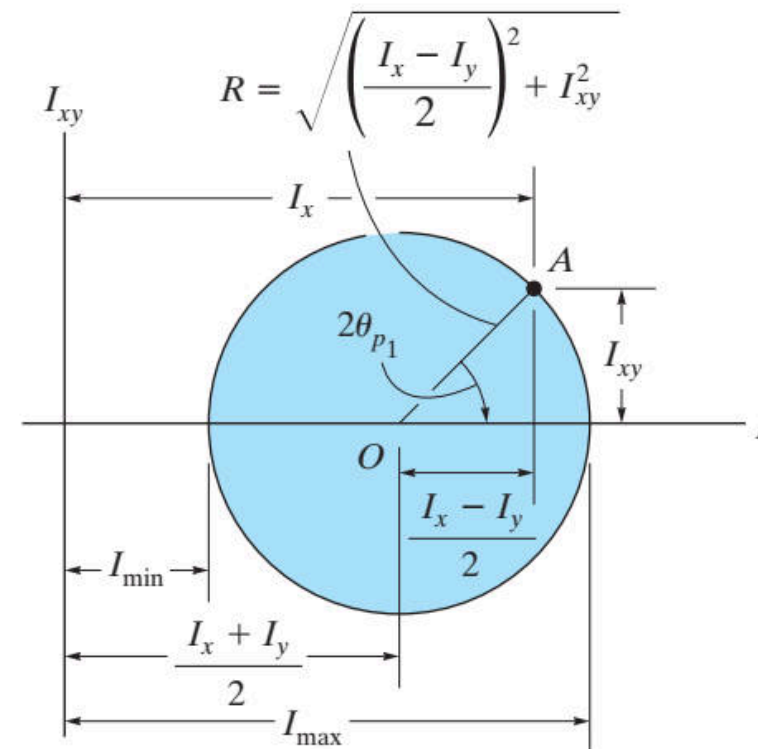
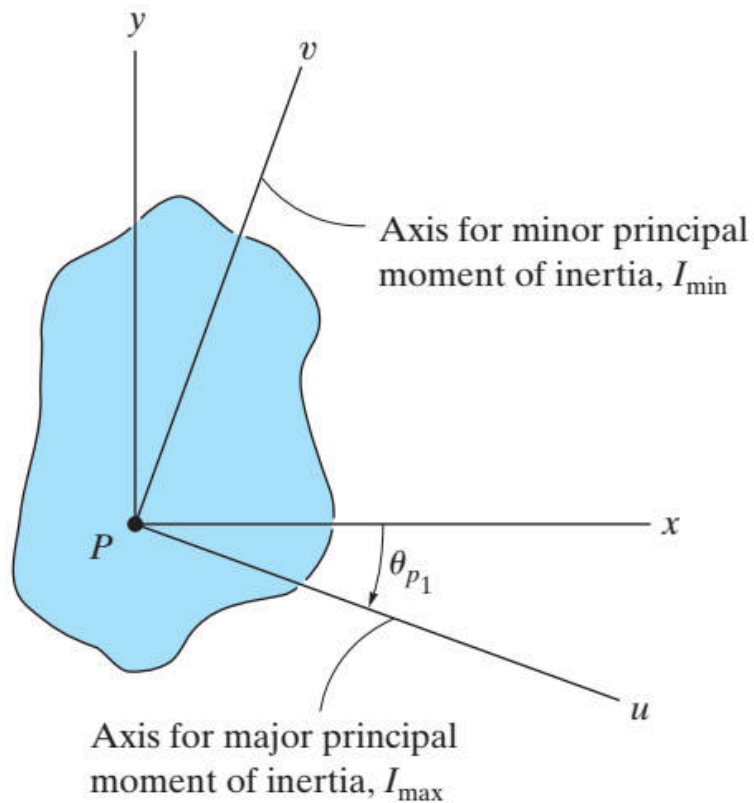
$$I_{uv} = \frac{I_x - I_y}{2} \sin 2\theta + I_{xy} \cos 2\theta$$

Squaring the first and third equation:

$$\left(I_u - \frac{I_x + I_y}{2} \right)^2 + I_{uv}^2 = \left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2$$

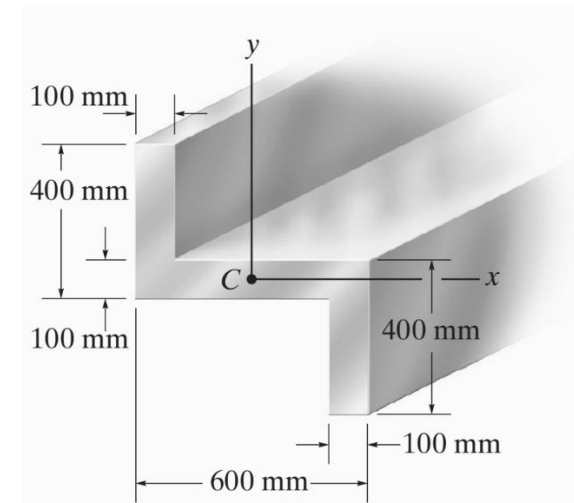
Mohr's Circle for Moments of Inertia

$$\left(I_u - \frac{I_x + I_y}{2} \right)^2 + I_{uv}^2 = \left(\frac{I_x - I_y}{2} \right)^2 + I_{xy}^2$$



Example: Determine the principal moments of inertia and the orientation of the major principal axes of inertia for the cross-sectional area of the member shown below, with respect to an axis passing through the centroid. Also check the eigenvalues and eigenvectors of the moments of inertia tensor.

$$I_x = 2.90(10^9) \text{ mm}^4, I_y = 5.60(10^9) \text{ mm}^4, \text{ and } I_{xy} = -3.00(10^9) \text{ mm}^4.$$



Practice at Home

1. Determine the directions of the principal axes with origin located at point O and the principal moments of inertia of the area about these axes.

